JUAN SEBASTIÁN SÁNCHEZ URBANO

Digital Solutions Developer, Machine Learning & Cloud (AWS) · Industrial Automation Engineer

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LinkedIn · GitHub · Portfolio

PROFESSIONAL PROFILE

I am driven by curiosity and the enjoyment of building things that truly work, whether prototyping a physical part on CNC or deploying a solution to the cloud with Artificial Intelligence. I am an Industrial Automation Engineer evolving toward Data Science and digital solutions development. I help companies and projects save time and scale through web development, intelligent automation (n8n, AI chatbots), Machine Learning and AWS services.

TECHNICAL SKILLS

Machine Learning & AI: Python (Scikit-Learn, XGBoost), Predictive modeling, Classification, AI agents, Conversational chatbots, NLP (TF-IDF, PCA), Cross-validation.

Cloud, Data & Automation: AWS (Serverless architecture), n8n (workflow automation), ETL pipelines, Medallion architecture (Bronze/Silver/Gold), REST APIs, Advanced SQL, MySQL.

Web, Apps & BI: HTML5, CSS3, JavaScript, Interactive dashboards, Power BI, R, Advanced Excel, Git, GitHub Actions (CI/CD), GitHub Pages, Bash, YAML.

Engineering & B2B consulting: Consultative sales (Sales Engineering), Industrial automation (PLC, CNC, Robotics, IoT), CAD/CAM prototyping and 3D printing, PDCA methodology.

PROFESSIONAL EXPERIENCE

Founder & Digital Solutions Consultant — JS Soluciones · Cali, Colombia

Sep 2026 – Present

- Design and deployment of custom digital solutions for businesses: web pages and applications, process automation with n8n, AI chatbot integration and cloud services (AWS).
- **Achievement:** Achievement: end-to-end management of the commercial and technical cycle, from requirements diagnosis to delivery in production.

Technical, Commercial and Data Automation Advisor — Proeléctricos · Popayán, Cauca

2019 – Nov 2025 · Mar – Sep 2026

- I analyzed and validated customers' technical and operational information to structure comprehensive solutions, ensuring the quality and traceability of commercial records.
- **Achievement:** Achievement: Designed an automated n8n pipeline connected to a web-channel chatbot that extracts, validates and routes customer reports and requests in real time to the responsible teams.

CNC and Robotics Area Leader (Production Analytics) — Metálicas Mundial · Candelaria, Valle del Cauca

Oct 2025 – Jan 2026

- I led the capture and analysis of critical manufacturing and quality-control variables, ensuring the reliability of process data.
- **Achievement:** Achievement: I identified bottlenecks and optimized cycle times through operational analytics and continuous improvement (PDCA), increasing the area's productivity.

Founder & Digital Manufacturing Designer — JS Taller · Cali, Colombia

2025 – Present

- Parallel rapid-prototyping venture: design, 3D printing and small-scale CNC router machining (acrylic cutting and technical materials) for custom solutions.
- **Achievement:** Achievement: custom 3D-printed products, shown in this portfolio's store and 3D-printing section.

FEATURED PROJECTS

Serverless engine on AWS and generative AI

AWS · Python · APIs · ETL · AWS, Python, APIs, ETL

- End-to-end automation to generate, process and publish digital content, optimizing cloud resources.
- **Impact:** Cloud resource optimization with serverless architecture.

Predictive maintenance with Machine Learning

Simulated data · Machine Learning, Classification, Python

- Fault-severity classification model built from machine signals, with cross-validation.
- **Impact:** Predictive model of fault severity for industrial machinery with 74% accuracy on 720 simulated signals, prioritizing critical faults.

SIVIGILA epidemiological ETL 2018–2022

Dengue · SARI · Tuberculosis · Python, Excel, Medallion, HTML dashboard

- Multi-page data extraction, transformation and loading (2018–2022) deployed as an interactive web dashboard.

EDUCATION

Complementary training

- Artificial Intelligence Bootcamp (MinTIC / Talento Tech) · Data Science, Python and Machine Learning path (Platzi) · AWS Cloud preparation Platzi profile

Industrial Automation Engineering

Universidad del Cauca · Graduated 2025